











Update the code coverage report periodically

Build/Produce Code (unit tests)

Deploy Product

Run the product
- Start code coverage tool

Run test suite

For each test

- Run test
- Store coverage data
- Feed it to metrics coverage
- Generate HTML report
- Process coverage data
- Show the mapping

Get feedback from your code changes

Code changes

Full Request

Get workflow for Test Selection Process

Run selected tests

Success

No

Stop

Run Success

Tests triggered every code change



Code triggered (Tests)

Block All Steps

Artifact
- Application logs
- Test report



Smart
Test
Selection

1. Map Tests to Code

- Periodically run tests with coverage tools (e.g., JaCoCo for Java).
- Store a map: Code (method/class) → Tests covering it (e.g., in Redis).

Key Benefits

Faster feedback loops

Efficient resource usage

Increase developer confidence

2. Analyse PR

- Identify exact code methods/classes modified in the Pull Request.

3. Select & Execute

- Query the map to find tests associated with the changed code.
- Run this smaller, targeted suite for faster, meaningful feedback.

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Synthetic Monitoring Through E2E Tests

